GREAT Cell Modelling Specification

DECOVALEX 2027 – SAFENET-2

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Abstract

This specification fixes geometry, material parameters, loads, boundary conditions and outputs for modelling the GREAT cell experiments, so that all DECOVALEX SAFENET-2 teams simulate the same problem and their results can be compared directly. It is code-independent. OpenGeoSys (OGS) reference solutions exist for the cases M1 and THM2.

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1 Purpose and scope

- **Fixed:** everything in Sections 3 to 8. Deviations must be reported.
- **Free:** numerical method, element type, mesh, time stepping and solver, within the accuracy targets of Section 9.
- **Units:** SI throughout (m, s, Pa, K, kg). Strains as fractions; plots may use 10^{-2} %.
- **Model dimension:** 2D plane strain, horizontal mid-section of the sample, unit thickness 1 m. All rates are per metre thickness.

2 Benchmark matrix

The benchmarks add one process at a time: mechanics (M), then fluid flow (HM), then heat (THM), each for an intact sample, a single fracture and a fracture network (Fig. 1, Table 1). This version specifies **M1** and **THM2** in full; the fractured cases follow the same conventions and are added later.

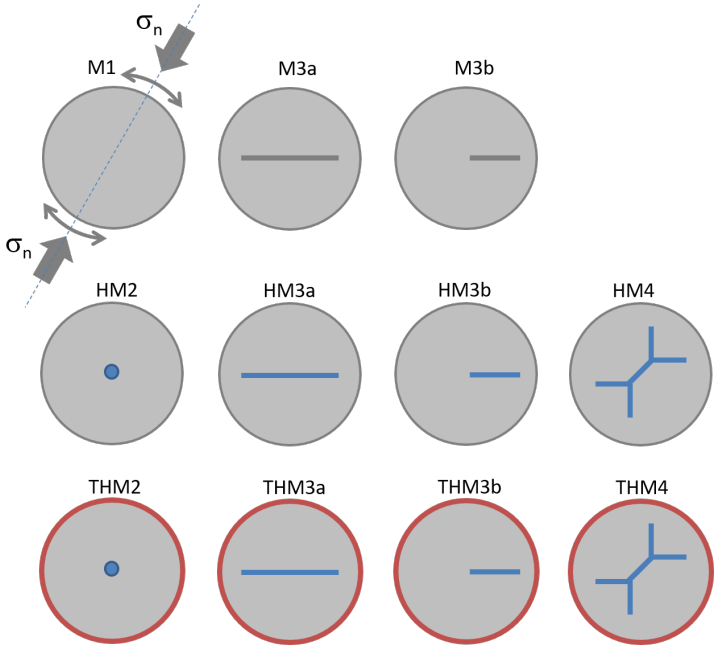


Figure 1: GREAT cell benchmark concept: M (grey), HM (blue, fluid injection) and THM (red boundary, heating) for intact, fractured and networked samples.

Teams are encouraged to run the stepwise sequence T → TH → THM used for the reference model, since each step has its own check.

3 Geometry and conventions

The domain is a disc of radius 0.100 m centred at $(0, 0)$: rock inside $r = 0.097$ m and a 3 mm rubber sheath outside it (Fig. 2, Table 2). Angles θ are measured counter-clockwise from the $+x$ axis.

The PEE and DSS arc widths (18°/4.5°) are estimates from Fig. 4b of [1]; teams should use them until measured values are available.

4 Mesh

The mesh is the team's choice, but it must resolve four features: the sheath, the PEE/DSS arc ends, the probe circle and the centre point.

- PEE/DSS arc ends and the two fixed points must be mesh nodes.
- The probe circle should be embedded as mesh edges, or outputs interpolated onto it at $\leq 1°$ spacing.
- At least two element layers across the 3 mm sheath.

Table 1: Benchmark cases.

Case	Sample	Processes	Status
M1	Intact	Poly-axial load, rotated (Loads A, E, B, F, C)	Specified; OGS reference
M3a / M3b	Full / half fracture	Mechanics	Geometry given; load as M1
HM2	Intact, central borehole	Load + injection	Covered by THM2 without heating
HM3a / HM3b	Full / half fracture	Load + injection into fracture	To be specified
HM4	Fracture network	Load + injection	To be specified
THM2	Intact	Load A + heating + cold-water injection	Specified; OGS reference
THM3a/b, THM4	Fractured	As HM3/HM4 + heating	To be specified

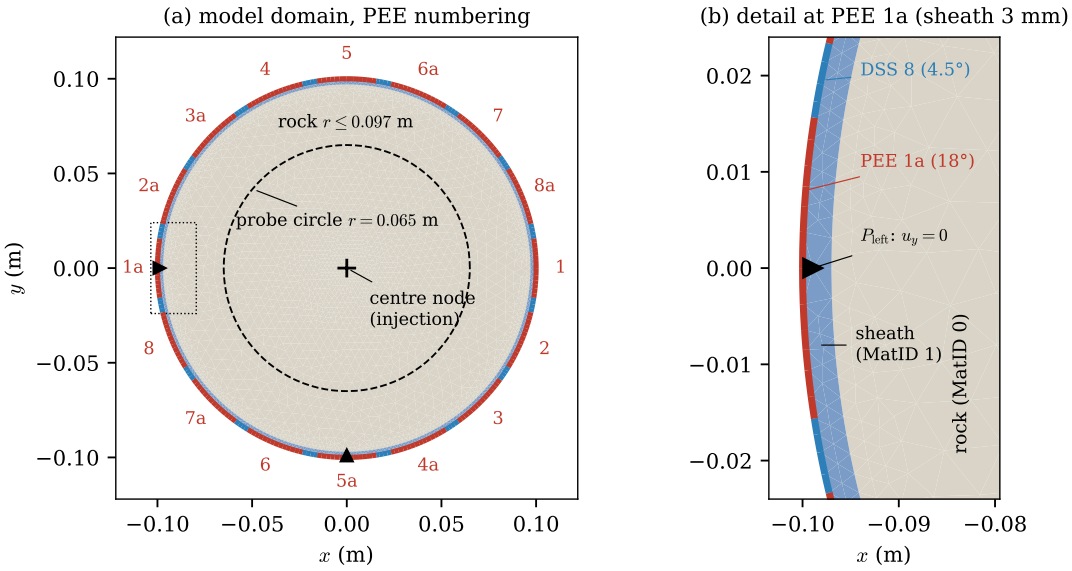


Figure 2: Geometry and PEE numbering (a), detail of the sheath at PEE 1a (b).

Table 2: Geometry and conventions.

Item	Specification
Rock	$r \leq 0.097\,\mathrm{m}$ (material 0)
Sheath	$0.097\,\mathrm{m} < r \leq 0.100\,\mathrm{m}$ (material 1)
Centre point	$(0, 0)$; injection point (THM2), borehole not resolved
Probe circle	$r = 0.065\,\mathrm{m}$; volumetric strain output
PEE arcs	16 arcs of 18° on $r = 0.100\,\mathrm{m}$. PEE i centred at $\theta = -(i-1)\cdot 22.5°$ ($i = 1 \ldots 8$, clockwise from $+x$); PEE ia opposite, at $+180°$
DSS arcs	16 arcs of 4.5° between the PEEs. DSS i lies between PEE i and PEE $i+1$; DSS 8 between PEE 8 and PEE 1a
Rigid-body control	$P_{\mathrm{bottom}} = (0,\ -0.09894)\,\mathrm{m}$: $u_x = u_y = 0$. $P_{\mathrm{left}} = (-0.09894,\ 0)\,\mathrm{m}$: $u_y = 0$
Full fracture (M3a/HM3a)	Segment $(-0.094,\ 0.094)\,\mathrm{m} \times \{0\}$
Half fracture (M3b/HM3b)	Segment $(0,\ 0.04)\,\mathrm{m} \times \{0\}$

- Refinement towards the centre for injection; the reference uses $h = 1.5\,\mathrm{mm}$ at the centre, 6 mm at $r = 0.065\,\mathrm{m}$ and 2 mm in the sheath.
- Reference mesh: 11 889 nodes, 5 804 quadratic triangles; displacement quadratic, pressure and temperature linear.

5 Material parameters

All materials are linear elastic and isotropic. Rock values follow Table 1 of [1]; hydraulic and thermal values follow the reference TH model (Table 3).

Table 3: Material parameters.

Parameter	Unit	Greywacke	Gneiss	Sheath
Young's modulus E	GPa	26.85	83.9	0.1
Poisson's ratio ν	–	0.27	0.21	0.4
Solid density ρ_s	$\mathrm{kg\,m^{-3}}$	2650	2750	1200
Porosity ϕ	–	0.005	0.001	as rock
Biot coefficient α_B	–	0.8	0.6	0
Permeability k	$\mathrm{m^2}$	1.409×10^{-14}	1.409×10^{-14}	$1.409 \times 10^{-14*}$
Bulk thermal conductivity λ	$\mathrm{W\,m^{-1}\,K^{-1}}$	2.8	2.8	2.8
Bulk heat capacity ρc	$\mathrm{J\,m^{-3}\,K^{-1}}$	2.133×10^6	2.133×10^6	2.133×10^6
Linear thermal expansion α_T (solid)	$\mathrm{K^{-1}}$	8×10^{-6}	8×10^{-6}	8×10^{-6}

* See open items (Section 10).

Water: $\rho_f = 1000\,\mathrm{kg\,m^{-3}}$ at $p_0 = 1\,\mathrm{MPa}$, compressibility $4.4 \times 10^{-10}\,\mathrm{Pa^{-1}}$, $\mu = 1 \times 10^{-3}\,\mathrm{Pa\,s}$, $c_f = 4180\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $\lambda_f = 0.6\,\mathrm{W\,m^{-1}\,K^{-1}}$, no thermal expansion. Storage follows from ϕ, α_B, the water compressibility and the elasticity. Bulk λ and ρc are prescribed; teams back-calculate solid values for their own mixing rule.

6 Mechanical load cases

Each load case prescribes a compressive normal stress on the 16 PEE arcs (Table 4 of [1]); PEE i and PEE ia carry the same value. Loads E, B, F and C are Load A rotated clockwise by 22.5°, 45°, 67.5° and 90° (Table 4).

Table 4: PEE normal stress per load case (MPa, compression positive).

Load	Rotation	PEE 1	PEE 2	PEE 3	PEE 4	PEE 5	PEE 6	PEE 7	PEE 8
A	0°	10.0	6.64	4.46	1.17	1.0	3.82	7.80	9.95
E	22.5°	9.95	10.0	6.64	4.46	1.17	1.0	3.82	7.80
B	45°	7.80	9.95	10.0	6.64	4.46	1.17	1.0	3.82
F	67.5°	3.82	7.80	9.95	10.0	6.64	4.46	1.17	1.0
C	90°	1.0	3.82	7.80	9.95	10.0	6.64	4.46	1.17

- **DSS:** each DSS carries the mean of its two neighbouring PEEs, $\sigma_{\mathrm{DSS},i} = (\sigma_{\mathrm{PEE},i} + \sigma_{\mathrm{PEE},i+1})/2$, with PEE 9 ≡ PEE 1.
- **Out-of-plane:** plane strain; no axial load is applied in 2D.
- **Gravity:** none (horizontal section).
- **Load path:** applied in the first time step and then held constant.

7 THM2 scenario

THM2 loads the intact sample with Load A, heats it from the outer boundary for 4000 s, injects cold water at the centre for 50 s and then shuts in until 5000 s (Table 5, Fig. 3). M1 is the same model with only the mechanical load, evaluated at the end of the first step.

Table 5: THM2 phases and boundary conditions.

Phase	Time (s)	Thermal	Hydraulic	Mechanical
0	$t = 0$	$T = T_0$	$p = p_0$	Load A applied
1 Heating	0–4000	Outer boundary: 293 → 363 K linearly over 0–325 s, then 363 K	$p = p_0$ on outer boundary	Load A held
2 Injection	4000–4050	As phase 1; centre point $T = 293$ K	Centre point: $Q = 1$ l/min per m $= 1.667 \times 10^{-5}$ m^3 s^{-1} per m (16.67 g s^{-1} per m)	Load A held
3 Shut-in	4050–5000	As phase 1	No injection	Load A held

Initial state: $T_0 = 293$ K, $p_0 = 1$ MPa, $\mathbf{u} = \mathbf{0}$ everywhere. Initial effective stress $\alpha_B p_0$ in the rock (0 in the sheath), so that total stress is zero before loading. Report strains relative to this state.

Boundaries: the outer boundary is drained ($p = p_0$) and has the prescribed temperature in all phases. The reference model treats the sheath hydraulically and thermally like the rock.

Time stepping (reference): 5 s to 400 s, 40 s to 4000 s, 1 s during injection, 2 s to 4150 s, 10 s to 5000 s. Teams may use other steps but must hit all output times exactly.

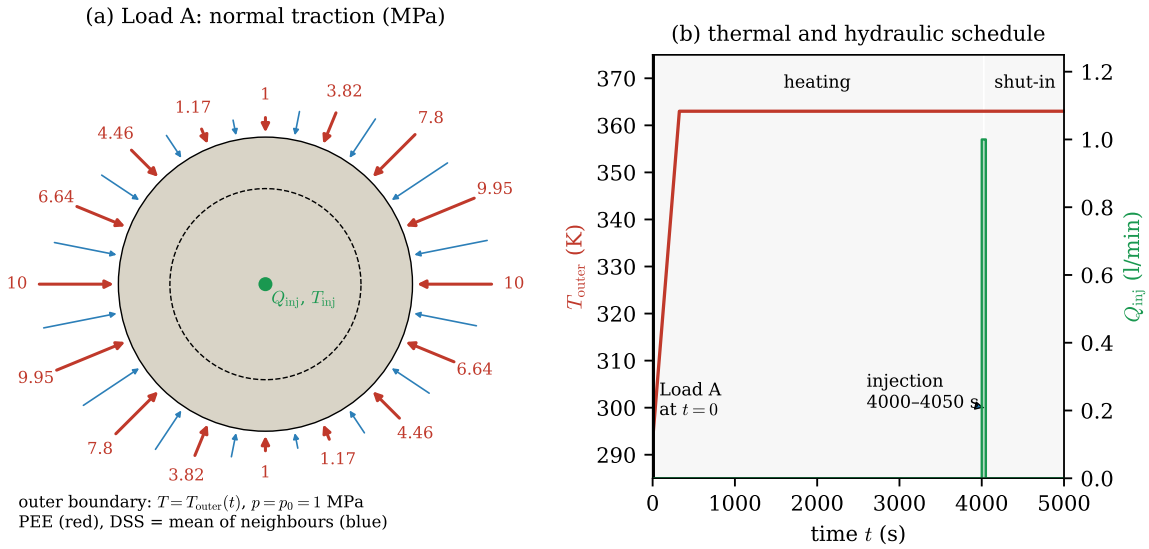


Figure 3: THM2 boundary conditions: Load A tractions (a) and the thermal and hydraulic schedule (b).

8 Required outputs

Each team delivers four CSV files per rock and load case, in SI units, comma-separated, with one header line (Table 6). File names follow `<team>_<case>_<rock>_load<X>_<output>.csv`, e.g. `UFZ_THM2_greywacke_loadA_probe.csv`.

Table 6: Required output files.

Output	Location	Times	Columns
`centre`	Centre point $(0, 0)$	Every time step, at least every 5 s; every 1 s in 4000–4060 s	`t, T, p`
`probe`	Probe circle $r = 0.065$ m, 360 points at 1°	Output times	`t, theta_deg, eps_xx, eps_yy, eps_xy, eps_vol, eps_tt, T, p`
`profile`	Radial line $\theta = 0°$, $0 \le r \le 0.1$ m, 101 points	Output times	`t, r, T, p, sigma_rr, sigma_tt`
`mean`	Probe circle average	Every time step	`t, eps_vol_mean, T_mean, p_mean`

Output times: 5, 325, 2000, 4000, 4001, 4005, 4010, 4025, 4050, 4060, 4100, 4200, 4500, 5000 s. For M1 only the state after loading is required ($t = 5$ s or the end of the mechanical step).

Sign convention: tension positive for stress and strain in the output files (the PEE loads above are given as compression positive). $\varepsilon_{vol} = \varepsilon_{xx} + \varepsilon_{yy}$; $\varepsilon_{\theta\theta}$ (`eps_tt`) is the circumferential strain, comparable with the fibre-optic measurement.

Teams also submit a short model description: code and version, process formulation, element type, mesh size and node count, time stepping, and any deviation from this specification.

9 Verification checks

Each team should pass the checks in Table 7 before submitting; reference values are from the OGS 6.5.9 model.

Table 7: Verification checks and reference values ($\varepsilon_{\mathrm{vol}}$ in 10^{-2} %).

Check	Quantity	Reference (OGS)	Independent check	Target
T: ramp only	$T(r,t)$	–	Duhamel solution, disc $R = 0.1$ m	≤ 0.2 K
Heating	Centre T at 4000 s	356.7 K	T model	≤ 0.5 K
Injection	$p - p_0$ at $r = 0.065$ m	0.0812 MPa	Thiem: 0.0811 MPa	≤ 1 %
Injection	$p - p_0$ at centre node	1.12 MPa	Mesh-dependent	Report only
Cooling	Centre T at 4100 / 5000 s	309.5 / 353.6 K	–	≤ 1 K
M1	Mean $\varepsilon_{\mathrm{vol}}$ on probe circle	-2.46 (Greywacke), -0.95 (Gneiss)	–	≤ 2 %
Heating	Mean $\varepsilon_{\mathrm{vol}}$ at 4000 s	11.15 (Greywacke), 12.03 (Gneiss)	$2(1+\nu)\alpha_T \Delta T + \varepsilon_{\mathrm{mech}}$, $\Delta T = 67.0$ K	≤ 2 %
Injection	Min. mean $\varepsilon_{\mathrm{vol}}$ after injection	10.73 at 4460 s (Greywacke), 11.62 at 4450 s (Gneiss)	–	Report
Recovery	Mean $\varepsilon_{\mathrm{vol}}$ at 5000 s	10.94 (Greywacke), 11.82 (Gneiss)	–	≤ 2 %

The strain on the probe circle varies with angle, with maxima near 100° and 280° under Load A. Temperature and pressure are the same for both rocks, because permeability and bulk thermal properties are equal.

10 Modelling choices and open items

Teams must state how they handle the following, since the specification leaves them open or the reference model uses a simplification:

- **Coupling scheme:** monolithic or staggered; convergence tolerances.
- **Mass vs. volume balance:** with pressure-dependent water density, injection is a mass rate ($16.67\,\mathrm{g\,s^{-1}}$ per m).
- **Point injection:** source at a node vs. resolved borehole; pressure at the injection point is mesh-dependent.
- **Thermal mixing rule** used to reach the prescribed bulk λ and ρc.

Open items for the task group:

☐ Sheath permeability: the paper gives $1 \times 10^{-19}\,\mathrm{m^2}$; the reference uses rock permeability to keep a drained outer boundary. Agree on one choice.

☐ Measured PEE and DSS arc widths (currently 18°/4.5° estimates).

☐ Injection rate per metre in 2D vs. the 0.2 m sample height in the experiment.

☐ Thermal expansion coefficient of the rocks ($8 \times 10^{-6}\,\mathrm{K^{-1}}$ assumed) and of water (neglected).

- ☐ Fractured cases (HM3, HM4, THM3, THM4): fracture properties, aperture law, injection into the fracture.
- ☐ Agree output formats and file naming proposed in Section 8.
- ☐ Comparison data from GREAT cell heating tests and fibre-optic strain.

References

[1] M. Mollaali, W. Wang, T. You, T. Nagel, A. Fraser-Harris, C. McDermott, and O. Kolditz. Numerical benchmarking of GREAT cell experiments: Poly-axial stress effects on fluid flow in fractured rock. *Journal of Rock Mechanics and Geotechnical Engineering*, 2026. doi: 10.1016/j.jrmge.2026.03.060.

[2] C. I. McDermott, A. Fraser-Harris, M. Sauter, G. D. Couples, K. Edlmann, O. Kolditz, A. Lightbody, J. Somerville, and W. Wang. New experimental equipment recreating geo-reservoir conditions in large, fractured, porous samples to investigate coupled thermal, hydraulic and polyaxial stress processes. *Scientific Reports*, 2018. doi: 10.1038/s41598-018-32753-z.

[3] O. Kolditz, C. McDermott, J. S. Yoon, J. Renner, L. Zhuang, A. Fraser-Harris, M. Chandler, S. Graham, J. Wang, and M. Mollaali. SAFENET-2 – fracture evolution in crystalline rocks (from lab to in situ scale). *Safety of Nuclear Waste Disposal*, 3:15–31, 2025. doi: 10.5194/sand-3-15-2025.